Sensitivity Analysis of Policy Relevant Treatment Effects to Failures of Monotonicity

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Abstract

The policy-relevant treatment effect (PRTE), understood as the local average treatment effect (LATE) in a different policy environment, addresses concerns of external validity for a study's conclusions. This paper provides a method for bounding PRTE without functional form or monotonicity, and interprets the PRTE as treatment effects of various subpopulations. The bounding framework uses the proportion of defiers relative to compliers as a sensitivity parameter and yields a convex identified set. The method is illustrated in an example where the same sex instrument is used to find the effect of having a third child on labor force participation.

Keywords: Instrumental variables, treatment effects, local average treatment effect, LATE, policy relevant treatment effect, PRTE, partial identification, monotonicity, sensitivity analysis

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1 Introduction

Researchers are often interested in the external validity of a study's conclusions, and the policy-relevant treatment effect (PRTE) is a benchmark for external validity. The PRTE is defined as the local average treatment effect (LATE) in a counterfactual environment. Consider the Angrist and Evans (1998) study that considers the effect of having a third child on the mother's labor supply. They used an indicator for whether the first two kids are of the same sex as an instrument for the third child since parents generally have a preference for gender balance among their children. Suppose the conclusion from the IV regression is that having a third child decreases the mother's employment probability by 0.083. This value is specific to the policy environment surrounding childcare in the dataset. Would we still have the same conclusion when the government gives a $5000 subsidy for childcare? What would the effect of a third child be for compliers in the new environment? Another example is from Duflo and Saez (2003), who are interested in the effect of a meeting (information treatment) on the take-up rate of a pension plan. They instrument for meeting attendance with a randomly assigned financial incentive. If we had given people $30 instead of $20 to attend the meeting, how different would the conclusion on the effect of the meeting be? More generally, with minimal assumptions, what can we say about external validity? What would the result of the experiment be if it had been designed differently? These are questions that are answered by the PRTE.[1]

Existing methods that target the PRTE rely on monotonic response to the instrument. Namely, procedures like Mogstad et al. (2018) rely on the marginal treatment effect (MTE) framework (Vytlacil (2002); Heckman and Vytlacil (2005)). MTE assumes an additively separable treatment selection equation, which is equivalent to monotonicity, the assumption that the instrument acts in the same direction for all individuals. However, monotonicity may not be a realistic assumption in many applications. For instance, in Angrist and Evans (1998), parents have a preference for gender balance among their children, so families with two boys or two girls are more likely to have a third child. But some parents may want two sons or two daughters, so they would violate monotonicity: under monotonicity, parents would have a third child if their first two children feature a balanced gender portfolio, while not having a third child if their first two children are of the same gender.

[1] Empirical examples that explicitly mention PRTE include Muralidharan et al. (2019) who are interested in the effect of an education program when implemented for 90 days when the experiment only ran for 86 days, Ito et al. (2021) who are interested in welfare effects under a counterfactual take-up incentive regime, and Carneiro et al. (2011) who are interested in the LATE for counterfactual education policies.

This paper proposes a method to place bounds on PRTE without monotonicity by leveraging a linear program similar to Mogstad et al. (2018).

The goal is to provide a method to bound the PRTE without the MTE framework, thereby accommodating monotonicity violations. To achieve this, I parameterize the problem using conditional means of potential outcomes for various subgroups characterized by their treatment response to the instrument so that bounds on objects of interest can be written as linear combinations of these parameters. The baseline specification of the constraint set is characterized by observed mean restrictions across conditional outcome distributions, but the method is amenable to additional restrictions the researcher may wish to impose. For sensitivity analysis to monotonicity violations, I use a sensitivity parameter that places an upper bound on the proportion of defiers (i.e., people who respond to the instrument in the opposite direction relative to the majority) relative to compliers (i.e., people who respond to the instrument in the same direction as the majority). The procedure then yields a convex identified set, and is illustrated in the Angrist and Evans (1998) example. This paper focuses on identification, so strategies for estimation and inference are only briefly mentioned.

The contribution of this paper is threefold. (1) This paper provides a tool for sensitivity analysis of monotonicity violations for various objects of interest, including the average treatment effect and LATE, through defier restrictions. This is especially useful for PRTE without functional form assumptions, which to the best of my knowledge does not yet have a viable method for sensitivity analysis. (2) The framework and theoretical results proved in this paper are more general than required for the PRTE. There is hence a theoretical contribution concerning the convexity of the identified set and sharpness of bounds generated by sensitivity analysis for a well-defined class of objects. (3) This paper interprets the PRTE in counterfactual policy environments without monotonicity. The PRTE literature relies on the MTE framework, so discussion and interpretation is based on a monotonic environment. The analogous PRTE objects without monotonicity and their interpretation remains unaddressed.

1.1 Literature Review

This paper relates to the literature that aims to relax the monotonicity assumption (e.g., DiNardo and Lee (2011); Heckman and Pinto (2018); Frandsen et al. (2019); Semenova (2020)), by proposing

an appropriate sensitivity parameter. The interpretation of the PRTE without monotonicity as the treatment effect of a subgroup of a population is in the spirit of De Chaisemartin (2017), which allows interpretation of IV estimates as the treatment effect on a subset of compliers.

Further, this paper contributes to the literature that puts bounds on (partially identify) objects of interest (e.g., Manski (1989); Balke and Pearl (1997); Horowitz and Manski (2000); Kitagawa (2021)). Narrower bounds are obtained by parameterizing the proportion of defiers as a sensitivity parameter. The method in this paper yields bounds in Balke and Pearl (1997) for the ATE when the outcome is binary. Noack (2021) conducted sensitivity analysis for the LATE. The approach proposed in this paper differs from that in that it nests LATE as a special case, and it uses a different sensitivity parameter i.e., the sensitivity parameter is proportion of defiers relative to compliers instead of the proportion of defiers.

Finally, the paper contributes to the growing literature on PRTE. Mogstad et al. (2018) used an additively separable selection equation in a linear program to obtain bounds on PRTE in the MTE framework, which assumes monotonicity, whereas my method treats monotonicity violation as a sensitivity parameter. There are also parametric approaches in Brinch et al. (2017) and Kline and Walters (2019) that are useful for obtaining policy effects. Parametric approaches are less useful when thinking of PRTE as a means to check external validity: since identification of heterogeneous treatment effects are often done without functional form assumptions, it is rather peculiar that functional form assumptions are imposed when we want to evaluate the robustness of the results.

2 General Framework for Partial Identification with Monotonicity Violations

This section presents the generic framework for sensitivity analysis for various treatment effects (TE). Policy-relevant treatment effects (PRTE) is a special case of the class of objects considered in this section. Subsections 2.1 and 2.2 adapt the setup in Mogstad et al. (2018) to accommodate a generic space of treatment response groups. Subsection 2.3 presents theoretical results for the general setup that are applicable to PRTE. An illustrative example with binary instrument is provided in Appendix B.1.

2.1 Setting

We observe random variables (T, Z, Y), denoting treatment, instrument, and outcome respectively. We are interested in the effect of endogenous T on Y. Outcome Y is unrestricted; instrument $Z \in \mathcal{Z} = \{0, 1, \cdots, k-1\}$ takes one of $k < \infty$ discrete values, and treatment $T \in \{0, 1\}$ is binary.[2] Let $T(z)$ denote the potential treatment when given instrument z, and let $Y(t)$ denote the potential outcome when given treatment t, which assumes that Y is not affected by Z directly. The instrument values are ordered such that $Pr(T(z) = 1)$ is increasing in z.[3] Then, the observed T and Y are given by:

$$T = \sum_{z \in \mathcal{Z}} 1[Z = z]T(z)$$
$$Y = \sum_{t \in \{0,1\}} 1[T = t]Y(t)$$

Treatment response groups G are characterized by the vector of potential treatments given the instrument, i.e., $G = (T(0), T(1), \cdots, T(k-1)) \in \mathcal{G}$.[4] Any two individuals in the same treatment response group thus have the same vector of potential treatments. We can denote the mass of each group in the population as

$$q_g := Pr(G = g)$$

Since there are 2 responses to each of the k instrument values, there are $d_q = 2^k$ groups. Using q to denote a vector that stacks all q_g values, the dimension of q is d_q. Similarly, define the conditional mean for each group as follows:

$$\mu_{gt} := E[Y(t)|G = g]$$

It is assumed that μ_{gt} is well-defined for any g with $q_g > 0$. Similarly, let μ stack all μ_{gt}, so its dimension is $d_\mu = 2^k \times 2$. The following assumption is maintained throughout:

[2] Strictly speaking, the setting is sufficiently general that it works for any discrete T. However, the binary T is a more natural way to think about TE.

[3] Due to Lemma 2 in Appendix D, there is a bijection from any set $\mathcal{Z}^*$ with k discrete values to $\mathcal{Z}$ such that for any $z, z' \in \mathcal{Z}$ such that $z > z'$, $Pr(T(z) = 1) \geq Pr(T(z') = 1)$. Hence, beyond having k discrete values for the instrument, assumptions on $\mathcal{Z} \subset \mathbb{N}$ and the ordering of the values are without loss of generality.

[4] This follows notation common in literature e.g., Blandhol et al. (2022).

Assumption 1. *(Independence). For all $t \in \{0,1\}$ and $z \in \mathcal{Z}$, $Y(t), T(z) \perp Z$.*

Random assignment of the instrument Z satisfies Assumption 1. In addition to Assumption 1, following Angrist and Imbens (1994), many papers also assume monotonicity, the assumption that the instrument works in the same direction for all individuals.

Assumption 2. *(Monotonicity). For all $z, z' \in \mathcal{Z}$, either $Pr(T(z) \geq T(z')) = 1$ or $Pr(T(z) \leq T(z')) = 1$.*

Assumption 2 implies that there are particular groups g with $q_g = 0$. This paper conducts sensitivity analysis for the failure of this assumption, so it relaxes Assumption 2. Since this assumption is a statement about the potential treatment response, sensitivity analysis involves careful consideration of the masses q_g of various treatment response groups.

2.2 Method Description

The method places bounds on objects of interest by using the researcher's input for a sensitivity parameter. To explain this, I first discuss objects of interest and constraints on μ implied by Assumption 1 in Section 2.2.1, where it is assumed that the vector q is known. Then, Section 2.2.2 shows how the divergence of q from monotonicity can be captured by a single sensitivity parameter.

2.2.1 Objects of Interest and Constraint Set

Objects of interest considered in this paper, including PRTE, can generically be written as:

$$\beta = \sum_{g,t} c_{gt}(q)\mu_{gt} = c(q)'\mu \tag{1}$$

where $c_{gt}(q)$ denote the weights on each of the μ_{gt}, and these coefficients can depend on q. The equality requires the object of interest to be linear in μ. $c(q)$ is the coefficient vector, with $c : [0,1]^{d_q} \to \mathbb{R}^{d_\mu}$ transforming the vector of proportions into weights on the conditional expectations. I require $c(.)$ to be continuous over q. Common objects of interest like LATE and the average treatment effect (ATE) can be written in this form. Further examples of the $c(q)$ vector are

provided in Table 2 of Appendix B.1. It suffices to mention for now that once q is known, $c(q)$ is known.

The μ object is constrained by the observable distribution in the data, and I use $\mathcal{M}(q)$ to denote the set of μ that satisfies defined equality and inequality constraints, and these constraints may depend on q. These constraints may arise from ex ante restrictions, or from the data. The researcher can specify what these constraints are, but I require equality and inequality constraints to be linear in μ and set $\mathcal{M}(q)$ be convex.[5]

While $\mathcal{M}(q)$ is flexible, I impose a set of constraints that I call "observed means" throughout the paper, because it is implied by Assumption 1. In $Y|T = t, Z = z$, the mean of the various μ_{gt} such that $T(z) = t$ weighted by their proportions has to be equal to the mean $E[Y|T = t, Z = z]$. Hence, for all z, t, these take the form:

$$\sum_{g:T(z)=t} q_g \mu_{gt} = p_{tz} E[Y|T = t, Z = z] \tag{2}$$

Observe that Equation (2) is a function of the q vector, so q parameterizes the constraint set $\mathcal{M}(q)$. Without ex ante restrictions, the set of μ that satisfies observed means is in Equation (3). This set is denoted $\mathcal{M}_m(q)$ to avoid confusion with the generic constraint set $\mathcal{M}(q)$.

$$\mathcal{M}_m(q) := \left\{ \mu \in \mathbb{R}^{d_\mu} : \sum_{g:T(z)=t} q_g \mu_{gt} = p_{tz} E[Y|T = t, Z = z] \qquad \forall z \in \mathcal{Z}, t \in \{0, 1\} \right\} \tag{3}$$

The constraint in set $\mathcal{M}_m(q)$ does not exploit all distributional information, but nonetheless makes the problem tractable and yields sharp bounds when the outcome is binary. As a default, if the researcher does not wish to impose additional restrictions, the method sets $\mathcal{M}(q) = \mathcal{M}_m(q)$. Additional restrictions that the researcher may impose include selection into treatment in the spirit of Roy (1951). Without binary outcomes, we might consider additional constraints implied by Assumption 1, such as the trimming bounds discussed in Appendix B.2.

Remark 1. *(Testable Implications). If the set $\mathcal{M}$ is empty, then the model is misspecified: when $\mathcal{M}$ is unrestricted, we cannot find μ that satisfies the constraints, which came from Assumption 1.*

[5] Since every equality constraint can be written as two inequality constraints, the constraint set can be written as a set of linear inequality constraints.

This falsifies Assumption 1. A notable feature is that the setup here tests for independence directly, whereas many existing tests do a joint test of those assumptions with monotonicity (or additive separability). This framework can also be used for other tests, including testing for selection bias.[6]

2.2.2 Sensitivity Parameter

To form a sensitivity parameter for Assumption 2, I first define compliers and defiers. With an unordered pair of instrument values (z, z'), there is a subset of individuals who have treatment response $T(z) < T(z')$, and another subset with $T(z) > T(z')$. Among individuals who change their treatment response at that instrument margin, there is a majority who move in one direction. The individuals who move with the majority are called compliers; individuals who move with the minority are defiers. To express this formally, for unordered pair (z, z'), define sets of defiers and compliers respectively as:

$$S^d_{(z,z')} := \{g : T(z) < T(z')\}$$
$$S^c_{(z,z')} := \{g : T(z) > T(z')\}$$

Since the pair is unordered, the two sets are defined by the condition that: $Pr(G \in S^d_{(z,z')}) \leq Pr(G \in S^c_{(z,z')})$, which translates to $z > z'$ in our setting where $Pr(T(z) = 1)$ is increasing in z.

Assumption 2 is equivalent to having no defiers, so the sensitivity parameter should control the proportion of defiers, which then affects the q vector. Hence, sensitivity parameter λ imposes the restriction that, for all pairs (z, z'),

$$\sum_{g \in S^d_{(z,z')}} q_g \leq \lambda \sum_{g' \in S^c_{(z,z')}} q_{g'} \tag{4}$$

As a special case, there is only one such restriction for binary Z. The researcher can also use a different sensitivity parameter for every pair (z, z') in nonbinary settings - this does not change the method, and merely increases the number of sensitivity parameters. When $\lambda = 0$, there does not exist any pair of instrument values for which there are defiers.

[6] Test for selection bias corresponds to the quantity $E[Y(0)|T = 1] - E[Y(0)|T = 0]$. (see Section 4.3 of Mogstad et al. (2018)) This can be done by writing the objective function appropriately.

We require proportion vector $q \in \mathcal{Q}(\lambda)$, where $\mathcal{Q}(\lambda)$ is the generic constraint set. As in the treatment of $\mathcal{M}(q)$, the researcher can specify additional restrictions, but I propose the minimal set of restrictions here. Namely, the proportions chosen must be consistent with the observed $p_{tz} := Pr(T = t|Z = z)$.[7] Assumption 1 implies $\forall t, z$,

$$\sum_{g:T(z)=t} q_g = p_{tz} \tag{5}$$

Hence, the minimal constraint set should satisfy Equation (4) and Equation (5), so it takes the form $\mathcal{Q}(\lambda) = \mathcal{Q}_m(\lambda)$, where

$$\mathcal{Q}_m(\lambda) := \left\{ q \in [0,1]^{d_q} : \sum_{g \in S^d_{(z',z'')}} q_g \leq \lambda \sum_{g' \in S^c_{(z',z'')}} q_{g'}, \sum_{g:T(z)=t} q_g = p_{tz}, \forall (z', z''), t, z \right\} \tag{6}$$

Observe that $\sum_g q_g = 1$ is implied by the condition that $\sum_{g:T(z)=t} q_g = p_{tz}, \forall t, z$. $\lambda \in [0, 1)$ by construction as $\lambda = 1$ is ruled out for any relevant instrument. Then, the optimization problems to obtain upper and lower bounds on the object of interest are:

$$\underline{\beta}_\lambda := \min_{q \in \mathcal{Q}(\lambda)} \min_{\mu \in \mathcal{M}(q)} c(q)'\mu \qquad \overline{\beta}_\lambda := \max_{q \in \mathcal{Q}(\lambda)} \max_{\mu \in \mathcal{M}(q)} c(q)'\mu \tag{7}$$

2.3 Theoretical Properties

The subsection presents the main identification result of the paper, that the generic optimization problem is convex, and is sharp for binary outcomes. The results are more general than what is required as they concern a class of optimization problems with particular properties and the IV setting is a special case of these results.

To examine theoretical properties, it is helpful to break the optimization problem into the inner

[7] With k instrument values, we have $k + 1$ equations and 2^k unknowns in the setup of Equation (5). There is 1 At group, 1 Nt group, $(k - 1)$ complier groups, and $2^k - (k + 1)$ defier groups. Hence $2^k - (k - 1)$ restrictions on defiers are required to identify q. In many applications, further restrictions can reasonably be imposed, usually through weaker alternative monotonicity assumptions: for instance, unordered monotonicity from Heckman and Pinto (2018) reduces the number of defier groups to $(k - 1)$.

and outer problems. Write the inner optimization problem as follows:

$$\underline{R}(q) := \min_{\mu \in \mathcal{M}(q)} c(q)'\mu \qquad \overline{R}(q) := \max_{\mu \in \mathcal{M}(q)} c(q)'\mu \tag{8}$$

These optimization problems are linear programs by assumption, given q. Define our identified set:

$$\mathcal{B}_\lambda = \{b \in \mathbb{R} : b = c(q)'\mu \text{ for some } \mu \in \mathcal{M}(q), q \in \mathcal{Q}(\lambda)\} \tag{9}$$

This formulation allows us to compute a linear program in an inner loop while optimizing over q in the outer loop. This is convenient because linear programs are cheap to implement. Formalizing the setup described in the previous subsection in Assumption 3, it can be shown that the sensitivity region is convex, the main result of this paper. Unsurprisingly, the minimal set of restrictions $\mathcal{M}_m(q)$ and $\mathcal{Q}_m(\lambda)$ satisfy the requirements of Assumption 3.[8]

Assumption 3. *The following hold:*

(a) For all $g \in \mathcal{G}$, if $q_g > 0$, then μ_{gt} is well-defined $\forall t \in \{0,1\}$, i.e., there exists $K < \infty$ such that $|\mu_{gt}| \leq K$.

(b) Restrictions in $\mathcal{M}(q)$ can be written as a system of linear inequalities in μ such that $\mathcal{M}(q) = \{\mu : A(q)\mu \leq b(q)\}$ is continuous in $(A(q), b(q))$, and $\mathcal{M}(q)$ is convex. Hyperparameters $A(q)$ and $b(q)$ of the linear program are continuous in q.

(c) $c(q)$ is continuous in q.

(d) $\mathcal{Q}(\lambda)$ is a nonempty convex set for all $\lambda \in [0,1)$.

Theorem 1. *(Convexity and Sharpness). Suppose Assumption 1 and 3 hold. Then, either $\mathcal{M}(q)$ is empty for all $q \in \mathcal{Q}(\lambda)$ and hence $\mathcal{B}_\lambda$ is empty, or the closure of $\mathcal{B}_\lambda$ is equal to the interval $[\underline{\beta}_\lambda, \overline{\beta}_\lambda]$, where*

$$\underline{\beta}_\lambda = \min_{q \in \mathcal{Q}(\lambda)} \underline{R}(q) \qquad \overline{\beta}_\lambda = \max_{q \in \mathcal{Q}(\lambda)} \overline{R}(q) \tag{10}$$

Further, if $Y \in \{0,1\}$ is binary, using $\mathcal{M}(q) = \mathcal{M}_m(q) \cap \{\mu \in [0,1]^{d_\mu}\}$ and $\mathcal{Q}(\lambda) = \mathcal{Q}_m(\lambda)$ yields a sharp identified set $[\underline{\beta}_\lambda, \overline{\beta}_\lambda]$.

[8] To see that $\mathcal{Q}(\lambda) = \mathcal{Q}_m(\lambda)$ satisfies convexity, take any two elements $q^0, q^1 \in \mathcal{Q}_m(\lambda)$ with $q^0 \neq q^1$. Form convex combination $q^* = \alpha q^0 + (1-\alpha)q^1$, with $\alpha \in (0,1)$. Taking the weighted sums of the constraints on q^0 and q^1, $q^* \in \mathcal{Q}_m(\lambda)$ is immediate.

The first part of the theorem claims that the identified set is convex. To show this result, I exploit properties of the inner linear program that allows continuity in the hyperparameters to show that $\bar{R}(q)$ is continuous in q. Then, using the fact that the image of a connected set formed by defier proportions is a connected set, it follows that for all $\beta \in [\underline{\beta}_\lambda, \overline{\beta}_\lambda]$, $\exists q \in \mathcal{Q}(\lambda)$ that yields that β.

In general, when using $\mathcal{M}_m(q)$, the bounds are not sharp.[9] The problem occurs because we only used information on the means across distributions, and we have not yet exploited all distributional information. To make sharp bounds, we should instead use restrictions on the entire distribution, which is an issue considered in Section A.1. Sharpness comes at the cost of linearity in the inner problem. Nonetheless, if our outcome space is binary, the bounds obtained from the procedure and the sensitivity region will be sharp, the result in the second part of the theorem. This theorem does not require Z or T to be binary. It is proven by constructing a distribution for binary Y given (q, μ) in the constraint set. The theorem implies that Equation (2) is a sufficient constraint set. Adding more information from the observed data cannot give us tighter bounds: if we do indeed get tighter bounds with some feature of the data, then it means that the original optimum (q, μ) chosen is not consistent with the data.

Sharpness in the binary case gives equivalence to Balke and Pearl (1997) for ATE and Noack (2021) for the LATE as special cases. The non-sharpness of the general program is similar to the generic framework presented in Mogstad et al. (2018), which did not allow for monotonicity violations. Hence, if we are interested in sensitivity analysis of PRTE in Mogstad et al. (2018) to monotonicity failure, the linear but nonsharp method is appropriate.

3 Identification of Policy Relevant Treatment Effects

This section discusses and interprets PRTE, which is understood as the LATE in counterfactual policy environments (e.g., in Heckman and Vytlacil (2005); Carneiro et al. (2011); Mogstad et al. (2018)). LATE in Angrist and Imbens (1994) is defined under monotonicity as the TE for the subpopulation who respond (i.e., change their treatment status) to the instrument, which is equivalent to the TE on compliers (TEC).[10] Since LATE was defined on a subpopulation with a particular

[9] Identified set B is sharp if, for all $\beta \in B$, there exists (q, μ) such that $c(q)'\mu = \beta$, and (q, μ) is permissible by the data, i.e., given parameter (q, μ), there exists a joint distribution $(\tilde{T}, \tilde{Z}, \tilde{Y})$ generated from the parameters such that the joint distribution is equivalent to the observed distribution.

[10] In the presence of defiers, the TE on the marginal population (TEM) and TEC in Table 2 are no longer equivalent.

treatment response status, it is more sensible to define it as the TEC when there are defiers present. Hence, I define PRTE as the TEC in the counterfactual environment. I emphasize that this is the effect of treatment T on the outcome Y in a different policy environment, *not* the effect of the policy on Y. The two subsections discuss PRTE where monotonicity fails, which is novel in literature.

I broadly consider two classes of counterfactual policies[11]:

1. Change the value of the instrument e.g., in Duflo and Saez (2003), people were randomly given a letter that gave them \$20 if they attended the meeting, but we could have given people \$30 instead. Since the financial incentive is the instrument, this corresponds to changing the instrument value. Researchers were interested in the effect of the meeting.

2. Monotonically change the probability of being treated e.g., subsidize childcare in the Angrist and Evans (1998) context: regardless of a couple's gender preference, the probability of having a third child increases. This will be referred to as the "threshold shifting" counterfactual as it is equivalent to shifting the threshold that the individual unobservable has to cross in order to be treated. Details are in Remark 2.

These two classes of policies are distinct. Changing the value of the instrument means some subpopulation may be incentivized into treatment while others incentivized away from treatment due to the effect of the instrument in our non-monotonic environment. In contrast, increasing the probability of being treated means everyone is weakly incentivized into treatment, so those who were previously treated must still be treated. The PRTE of changing instrument value (Section 3.1) answers questions of the form "For an experiment that randomly assigns the instrument, what would the LATE be if the experiment had been designed differently?" The PRTE of threshold shifting (Section 3.2) answers questions of the form "What would the LATE be if we had a different policy environment where everyone is more (or less) likely to take up treatment given their instrument?"

At a high level, the PRTE aims to find the TEC in a policy environment different from what we had data for. While we have not seen people respond to this incentive, we have seen people respond

[11] Recent literature in PRTE as in Carneiro et al. (2010), Carneiro et al. (2011) and Mogstad et al. (2018) consider three counterfactual policies. These policy counterfactuals are in the class considered by Heckman and Vytlacil (2005), which involves policies that do not affect the marginal treatment response of T on Y. Their policy counterfactuals include (i) Additive α change in propensity score with the same instrument value (ii) Proportional $1+\alpha$ change in propensity score with same instrument (iii) Additive α shift of the jth component of Z, so $Z^* = Z + \alpha e_j$ and $p^*(x,z) = p(x,z)$. Changing the value of the instrument corresponds to policy type (iii) and monotonically changing the probability of being treated corresponds to (i) and (ii), so I group the first two together.

to other incentives. If we are willing to put bounds on the fraction of people who respond to the new but not the old, we can make progress. To bound such fractions, some economic reasoning is required for how the environment maps to the fraction: in Duflo and Saez (2003), we require a mapping from the financial incentive to fraction of people changing their behavior; in Angrist and Evans (1998), we require a mapping from the subsidy amount to the fraction.

Binary instrument will be used throughout this section for ease of exposition. Non-binary instrument is discussed in Appendix A.2. Since the PRTE is the LATE in the counterfactual environment, this translates to identifying the TE for a subgroup of the population, which is useful for assessing the robustness of conclusions on TE. The PRTE problem can be addressed by the framework presented in Section 2, so the result of convexity and sharpness in Theorem 1 apply. To implement the method, the researcher simply has to input the sensitivity parameter and the extrapolation parameter (which will be clarified in the subsections). In the binary setting, $T \in \{0,1\}$ and $Z \in \{0,1\}$. Since $k = 2$, $d_q = 4$ and $d_\mu = 8$. Let $T_g(z)$ denote the potential treatment $T(z)$ of group g when given instrument z. Using terminology in the literature, the 4 response groups are always-takers (A) with $T_A(0) = T_A(1) = 1$, compliers (C) with $T_C(0) = 0$ and $T_C(1) = 1$, defiers (D) with $T_D(0) = 1$ and $T_D(1) = 0$ and never-takers (N) with $T_N(0) = T_N(1) = 0$.

Remark 2. *(Relation to PRTE in MTE framework). Under Assumption 2, Mogstad et al. (2018) have a separable threshold-crossing model of the form $T = 1[\tilde{\nu}(Z) - u \geq 0]$, where u is the individual unobservable and one can innocuously normalize $u \sim U[0,1]$ such that low values of u get treated. But once we relax Assumption 2, separability fails and the latent threshold-crossing selection model is nonseparable, so $T = 1[\nu(Z,u) \geq 0]$. We should think about the ν function theoretically because it is not identified. The counterfactual policies thus map to (1) Change the value of the instrument so $T^* = 1[\nu(Z^*,u) \geq 0]$; (2) Change the threshold for everyone so $T_* = 1[\nu(Z,u) \geq -\alpha]$.*

3.1 Changing Instrument Value

Suppose we have a new instrument value Z^*. For illustration, I extrapolate the instrument rightward: in the original study, we have $Z \in \{0,1\}$, but now we have $Z^* = 2$. The reasoning is similar if we wish to interpolate the instrument, or extrapolate leftward.[12] For every original group

[12] For a concrete example of interpolation, consider Dupas (2014), who conducted an RCT with randomly assigned prices for an antimalarial bed net (Olyset net), so Z is the randomly assigned price, T is whether the household purchases the net, and Y is whether the household uses the net. The experiment had a range of prices from 0 to

Table 1: Classification of groups for various counterfactual policies. Column 2 presents the stem group, which is the original groups before extrapolation. Column 3 presents the mixture of groups for extrapolating rightward and Column 4 presents the mixture of groups under the threshold crossing counterfactual where the probability of treatment monotonically increases for everyone.

(T,Z) (1)	Stem Group (2)	Right Extrap (3)	Inc Prob TC (4)
T=0, Z=0	N, C	(0,0,0), (0,0,1), (0,1,1), (0,1,0)	CC, CA, NN, NC, ND, NA
T=1, Z=0	A, D	(1,1,0), (1,1,1), (1,0,0), (1,0,1)	DD, DA, A
T=0, Z=1	D, N	(1,0,0), (1,0,1), (0,0,0), (0,0,1)	DD, DA, NN, NC, ND, NA
T=1, Z=1	A, C	(1,1,0), (1,1,1), (0,1,0), (0,1,1)	A, CC, CA

$g \in \{A, C, D, N\}$, individuals can have two possible responses at $Z^* = 2$, resulting in 8 treatment response groups, given by $(T(0), T(1), T(2))$. For each triple $(T(0), T(1), T(2))$, I define its stem group as its original group. For instance, N is the stem group for $(0, 0, 0)$ and $(0, 0, 1)$ when extrapolating rightward, as both their original responses are $(0, 0)$, corresponding to N.

The four observed distributions will now each be a mixture of four extrapolated groups. Namely, in $T = 0, Z = 0$, we originally had N and C. When extrapolating rightward, since N is the stem group of (0,0,0) and (0,0,1), and C is the stem group of (0,1,1) and (0,1,0), the distribution $Y|T = 0, Z = 0$ now contains a mixture of four groups (0,0,0), (0,0,1), (0,1,1) and (0,1,0). The reasoning is analogous for other (T, Z) pairs, and the mixture of groups is presented in Table 1. Given q, $\mathcal{M}(q)$ is well defined by observed means as before.

PRTE is the TE for the "compliers" in the counterfactual environment. Since LATE is defined on an instrument pair, we have to consider which instrument pair the researcher is referring to. In the right extrapolation exercise, one of the instrument values has to be $Z = 2$ that we do not have data for, so the PRTE can be defined either at the $(0, 2)$ pair or the $(1, 2)$ pair. In the Duflo and Saez (2003) running example, we ask what the LATE of the experiment would have been if we had given people \$30 instead of \$20. This corresponds to the $(0, 2)$ pair, because the control group still did not receive any financial incentive, and the treatment group simply received a larger incentive. In the right-extrapolation setup, compliers are those who switch their treatment status from 0 to 1 at the 0-2 instrument margin. This would be groups (0,0,1) and (0,1,1). Thus, $PRTE = E[Y(1) - Y(0)|G \in \{(0, 0, 1), (0, 1, 1)\}]$. We can write this using the $c(q)'\mu$ notation for

250 Kenyan Shillings (Ksh), but I will stick to the binary case of 0 and 250 Ksh for illustration. The counterfactual policy that Mogstad et al. (2018) considered is offering the net at 150Ksh, which is the actual market price when the nets were rolled out a year later. If the experiment had been designed such that some had to pay 150Ksh and others got it free, the PRTE is the LATE at the 0-150 margin.

the objective function.[13]

$$PRTE = E[Y(1) - Y(0)|G \in \{(0,0,1),(0,1,1)\}] = c(q)'\mu = \sum_{g,t} c_{gt}(q)\mu_{gt}$$

It remains to consider what $\mathcal{Q}(\lambda)$ is with extrapolation. To decide what the appropriate extrapolation parameter is, the researcher should consider the groups of interest G^* (i.e., G whose TE features in the object of interest). Let S_G denote the stem group of G. Then, the extrapolation parameters are $Pr(G^*|S_{G^*})$ for all $G^* \notin \{G : Pr(G|S_G) = 1 \text{ under monotonicity}\}$. In this environment, our G^* are $(0,0,1)$ and $(0,1,1)$. Since $Pr((0,1,1)|C) = 1$ under monotonicity, the only extrapolation parameter is $Pr((0,0,1)|N)$. This set of extrapolation parameters are chosen so that extrapolation environment will be the same both with or without monotonicity.

At the 0-1 and 1-2 instrument margins, sensitivity restrictions are:

$$\begin{aligned} q_{(1,0,0)} + q_{(1,0,1)} &\leq \lambda(q_{(0,1,1)} + q_{(0,1,0)}) \\ q_{(0,1,0)} + q_{(1,1,0)} &\leq \lambda(q_{(0,0,1)} + q_{(1,0,1)}) \end{aligned} \tag{11}$$

With the extrapolation parameter, sensitivity restrictions, and data restrictions on p_{tz}, the constraint set for $\mathcal{Q}(.)$ can be written as $\mathcal{Q} = \mathcal{Q}_r(\lambda; Pr((0,0,1)|N))$, where

$$\mathcal{Q}_r(\lambda; Pr((0,0,1)|N)) = \left\{ q : \text{Equations (5) and (11) hold, and } \frac{q_{(0,0,1)}}{q_{(0,0,1)} + q_{(0,0,0)}} = Pr((0,0,1)|N) \right\} \tag{12}$$

Extrapolation is characterized by q, so the analysis here does not depend on the value of Z. Regardless of whether the new Z is 1.1 or 100, the same argument from extrapolating rightward applies. Instead, the approach parameterizes the extent of extrapolation by the q vector. Namely, $Z = 1.1$ is an environment that is very similar to the original policy, so we expect $Pr((0,0,1)|N)$ close to zero. In contrast, with $Z = 100$, or a very different propensity score, it is analogous to a

[13] The interpretation for the PRTE is the TE of the meeting for people who are somewhat sensitive to financial incentives: this group includes (0,0,1) who are less sensitive to the incentive than the original compliers who are (0,0,1). The coefficient takes the form:

$$c_{gt}(q) = \begin{cases} \frac{(-1)^{1-t} q_{(0,0,1)}}{q_{(0,0,1)} + q_{(0,1,1)}} & \text{if } g = (0,0,1) \\ \frac{(-1)^{1-t} q_{(0,1,1)}}{q_{(0,0,1)} + q_{(0,1,1)}} & \text{if } g = (0,1,1) \\ 0 & \text{otherwise} \end{cases}$$

large extrapolation with $Pr((0,0,1)|N)$ close to 1. For instance, this could be a monetary incentive, so having a large incentive would move all N into taking up treatment.

Remark 3. *Once PRTE is defined as the TE on some subpopulation, we can use the same method to obtain TE on other subpopulations that are potentially more interesting. For instance, (1,0,1) is the group that are defiers at the 0-20 margin, and compliers at the 20-30 margin, where 0, 20 and 30 are the amount of money offered to attend the meeting in Duflo and Saez (2003). Behavioral studies on fund raisers in Gneezy and Rustichini (2000) show such behavior exist, where giving a bit of financial incentive disincentivises intrinsic effort, but offering a large financial incentive increases their effort. PRTE answers: For people with such behavioral responses, what is their take-up rate of a pension plan?*

Remark 4. *(Overpowering Experiments). Having a large incentive, say $100 in the Duflo and Saez (2003) experiment, can incentivize many people into treatment (the meeting). But this also includes people who go just for the money rather than because they are interested in the pension plan. If the incentive were $5 instead, the LATE of information on taking up the pension plan is likely larger, since this excludes the people who are not interested in the plan to begin with. Exercise in extrapolation puts bounds on what the results of the experiment would have been if it had been designed differently.*

3.2 Changing Threshold Crossing Rule

Suppose we have a policy that incentivizes treatment for everyone.[14] In our policy counterfactual, always-takers will still be always-takers. Compliers can remain compliers, or they can be come always takers when the policy is strong enough to shift their $Z = 0$ treatment to $T = 1$. Same argument can be made for defiers. Finally, consider the never-takers. If the policy is weak, they would remain N. The policy may affect the outcome for only either $Z = 0$ or $Z = 1$, which changes their response behavior to D or C. The policy may also be strong enough to get the N group to $T = 1$ regardless of the instrument. Then, N can change their behavior to N, C, D, or A.

Use notation $g = GG'$ to denote the subset of the population who are in group G in the original dataset and group G' in this counterfactual policy. As before, each (T, Z) pair contains a mixture of response groups, as denoted in Table 1. The $PRTE$ is the TE on the new group of compliers.

[14] If the policy monotonically decreases treatment propensity for everyone, the reasoning is analogous.

This includes all response groups with $G' = C$, so PRTE is expressed as

$$PRTE = E[Y(1) - Y(0)|G \in \{CC, NC\}]$$

Here, we have two extrapolation parameters $Pr(CC|C), Pr(NC|N)$ which control the proportion of the old C and N groups going into the subpopulation that we are calculating an ATE for. To see how this follows from our general principle of choosing extrapolation parameters, our groups of interest G^* are CC and NC. Under monotonicity, stem group C can become CC or CA, and stem group N can become NC, NA, ND, or NN. Hence, for both G^*, $Pr(G^*|S_{G^*}) = 1$ is not implied by monotonicity. A policy environment that is very similar to what we have data for would have large $Pr(CC|C)$ and small $Pr(NC|N)$, as the proportion with $G' = G$ is large.

Although we still have a single pair of instrument values in this setting, they occur under different policy environments. Then, pairs of defier restrictions are:

$$\begin{aligned} q_{DD} + q_{DA} &\leq \lambda(q_{CC} + q_{CA}) \\ q_{DD} + q_{ND} &\leq \lambda(q_{CC} + q_{NC}) \end{aligned} \tag{13}$$

The threshold-crossing and instrument-value environments have different sources of uncertainty. In the (right-extrapolation) instrument-value setup, the uncertainty is in expected outcomes when $Z = 2$, while the data identifies expected outcomes for $Z = 0$ and $Z = 1$. In the threshold-crossing setup, the uncertainty is in the expected outcomes for both $Z = 0$ and $Z = 1$ under the new policy, which is not in the data. Even if we impose some monotonicity such that people treated in $Z = 1$ cannot be untreated in $Z = 2$ (thereby ruling out (0,1,0) and (1,1,0)), we only reduce the problem to 6 response groups, which is distinct from the threshold-crossing problem where propensity score is uniformly increased, yielding 9 response groups.

4 Implementation

To implement this method, we can simply plug in the sample analog. The researcher has to input four things: (i) data (T,Z,Y) (ii) object of interest (iii) extrapolation parameter q^e (if targeting PRTE) (iv) sensitivity parameter λ (if doing sensitivity analysis). The vector of μ's will be pa-

rameters that we optimize over in the inner loop of Equation (8). Let n denote the number of observations. An implementable algorithm follows:

1. Estimate probability objects p_{tz} by $\hat{p}_{tz} = \frac{\sum_{i=1}^{n} 1[T=t,Z=z]}{\sum_{i=1}^{n} 1[Z=z]}$. Use sample analog $\hat{E}[Y|T=t,Z=z] = \frac{1}{n_{tz}} \sum_{i:T_i=t,Z_i=z} Y_i$ for $E[Y|T=t,Z=z]$.

2. For given $q \in \mathcal{Q}(\lambda; q^e)$,

   (a) Plug in $\hat{E}[Y|T=t,Z=z]$ and q into Equation (2).

   (b) Set up the objective function and solve the linear program in Equation (8). Output the value of the objective function $R(q)$.

3. For given (λ, q^e), optimize output of Step 2 over q in the outer loop as in Equation (10) using the sample analog.

This procedure can estimate the bounds consistently, a result stated in Theorem 2 in Appendix C.1. A potential issue with this procedure is that the optimization problem over q in the final step can get unwieldy with a large number of sensitivity restrictions and a high-dimensional q. Nonetheless, due to the way the sensitivity parameter is defined, the $\binom{k}{2}$ sensitivity restrictions can be reduced to $k-1$ restrictions, due to Theorem 3 in Appendix C.1. To address the problem of a high-dimensional q, the optimization problem can often be simplified when we are only interested in the TE of a few groups. The dimension-reducing procedure for the empirical application is detailed in Appendix C.2, where, while having 9 treatment response groups, one can essentially optimize over 6 groups. For inference, it is possible to rewrite the constraints as a moment inequality model, which allows one to do inference using generalized moment selection, so procedures like Andrews and Soares (2010) can be implemented. More details are provided in Appendix C.3.

5 Empirical Application

Considering the Angrist and Evans (1998) problem, we are interested in the effect of a third child (T) on women's labor force participation (Y), and the instrument is whether the first two kids are the same sex (Z). All variables are thus binary. Defiers are parents who have a preference for either two boys or two girls. Following Angrist and Evans (1998), I focus on the 1990 PUMS data for

mothers using their posted dataset. The implementation follows their Table 5 where no additional covariates were included.

We have $n = 380007$ observations and the proportions are given by $\widehat{Pr}(Z) = 0.504$, $\widehat{Pr}(T = 1|Z = 1) = 0.402$ and $\widehat{Pr}(T = 1|Z = 0) = 0.339$. Hence, the first stage is 0.063. Sensitivity analysis bridges the two extremes of monotonicity and worst-case bounds for PRTE, giving bounds at intermediate values of λ. The bounds also vary continuously with λ. The key takeaway is that the monotonicity assumption is not innocuous: without that assumption, the bounds on LATE and PRTE can become uninformative.

5.1 Interpreting the PRTE

When considering counterfactual policy environments, the threshold shifting counterfactual in Section 3.2 is more sensible (e.g., childcare subsidy) than changing the instrument value. Then, PRTE is TE for compliers in the new policy regime, which are groups $\{CC, NC\}$ i.e., $PRTE = E[Y(1) - Y(0)|G \in \{CC, NC\}]$. These are people who used to be compliers and remain compliers, and people who were never-takers but become compliers when there is a childcare subsidy.[15]

The extent of extrapolation is characterized by proportions $Pr(CA|C), Pr(NC|N)$. Based on the principle of choosing extrapolation parameters in Section 3, we should have chosen $Pr(CC|C), Pr(NC|N)$ as our extrapolation parameters. However, we can make a simple transformation to $Pr(CA|C) = 1 - Pr(CC|C)$ to use $Pr(CA|C)$ as the extrapolation parameter instead. These proportions still determine the proportion of the original C and N groups switching to our groups of interest, and in the special case of $Pr(CA|C) = 0, Pr(NC|N) = 0$, the PRTE is simply the LATE of the original environment i.e., $E[Y(1) - Y(0)|C]$, which makes these objects more interpretable.

To implement the method, the researcher has to provide input for two questions: (1) What is the new policy environment? (2) How does this new environment map to $Pr(CA|C), Pr(NC|N)$? Since these are switchers in light of the policy, we would expect a \$2000 subsidy to yield larger proportions than a \$10 subsidy. PRTE then answers the question of what the LATE would be in this new regime.

[15] CC have very strong preference for gender balance, and are hence unmoved by the subsidy, and the NC group may be interpreted as those with weak preference for gender balance, but need a sufficient financial incentive to have a third child.

In this application, there is no natural benchmark in literature as to what these extrapolation parameters should be. Existing papers (e.g., Carneiro et al. (2010)) calculate a threshold-crossing PRTE at the margin i.e., $Pr(CA|C) \to 0, Pr(NC|N) \to 0$, so we may choose small values of these parameters to mimic that approach. I implement the procedure with $Pr(CA|C) = 0.05$ to exaggerate the difference between the curves in Figure 1 - reducing $Pr(CA|C)$ merely brings the curves closer to the original LATE.

Remark 5. *(Advantage of PRTE Setup). Setting up the PRTE without the MTE framework provides an explicit way of thinking about the subgroup of people who respond in the new policy environment. This is advantageous when calculating nonparametric bounds beyond the margin.*

This differing environment could be a difference place or time. Since the study was done on US data, if we believe that the Canadian population is similar to the US, and its only difference is that it has better childcare, then the PRTE is what the LATE in Canada would be. Another form of extrapolation is over time. The study used 1990 data, but the current policy environment has changed since then, so the PRTE tells us what the LATE would be now. Thus, minimally, PRTE is useful for robustness checks of the LATE.

The PRTE is as useful to the policy maker as the LATE. If LATE features in the policy function, then so must the PRTE once the policy is implemented because the environment would have changed. The complier group is environment-dependent. For example, if the policy maker wishes to give a \$2000 subsidy in two tranches, once the first \$1000 has been rolled out, the "LATE" would have changed, and we cannot expect the second \$1000 to yield the same effect. This occurs because people no longer stick to their original groups. Nonetheless, the method proposed here can put nonparametric bounds on the new LATE, which is the PRTE. Such a setting is relevant when policy makers only have old studies or surveys available to inform current policy implementation.

To clarify what the PRTE is, it is helpful to mention another interesting object that the PRTE is *not*. The policy effect (PE) is an object distinct from the PRTE, which is also potentially interesting. Suppose we are interested in expected outcomes Y for the new policy environment $Pol(1)$. The effect of the new policy can be written as $PE = E[Y|Pol(1)] - E[Y|Pol(0)]$. Here, "What is the impact on labour force participation of women with a \$2000 childcare subsidy?" is a question on PE, while "What is the impact of the third child women's labor force participation for women who

weakly prefer a balanced gender portfolio?" is a question on PRTE.[16] The expected outcome in the original setting is a linear combination of potential outcomes. Where $p_z := Pr(Z = 1)$, PE is:

$$PE = q_{CA}(1 - p_z)(\mu_{CA1} - \mu_{CA0}) + q_{DA}p_z(\mu_{DA1} - \mu_{DA0}) + q_{NA}(\mu_{NA1} - \mu_{NA0})$$
$$+ q_{NC}p_z(\mu_{NC1} - \mu_{NC0}) + q_{ND}(1 - p_z)(\mu_{ND1} - \mu_{ND0})$$

Observe that groups $\{A, CC, DD, NN\}$ do not feature in PE, since their behavior in response to the instrument is the same in both environments, so their outcomes are unaffected by the policy. This PE object is thus still a linear function of μ, and can be solved using the procedure described, though it is a simpler object to deal with.

5.2 Results

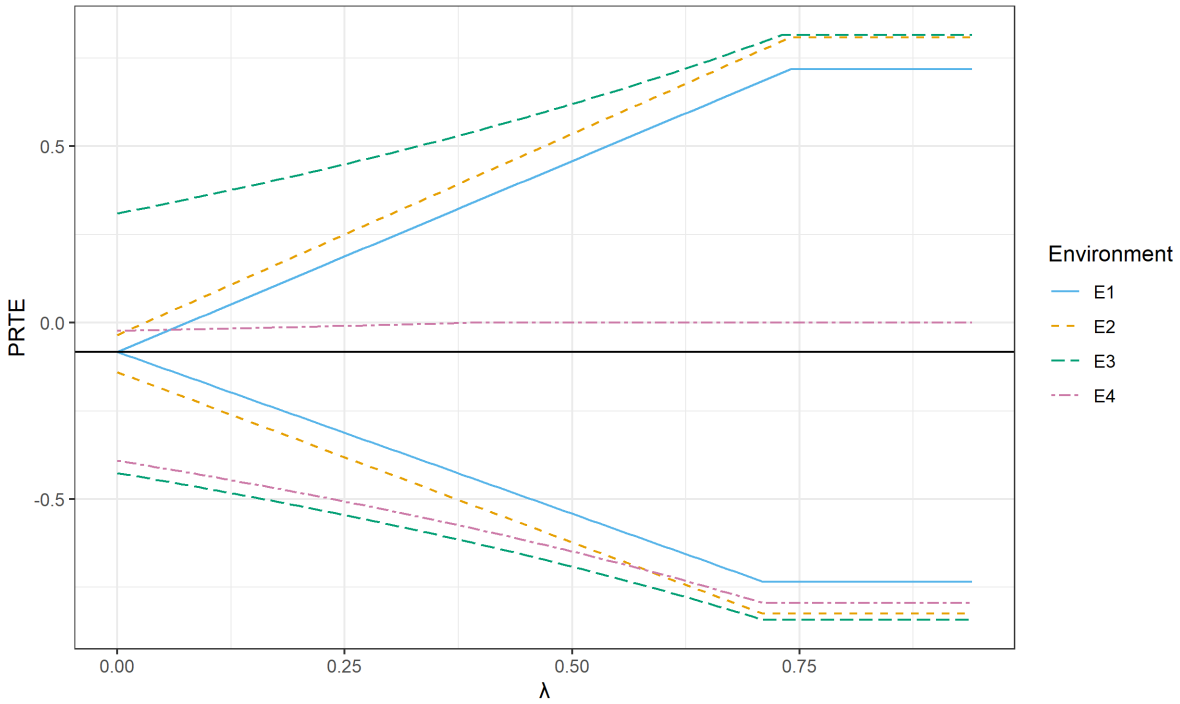
In this empirical illustration, I consider two counterfactual environments characterized by differing values of the extrapolation parameters $Pr(CA|C), Pr(NC|N)$. The main result for sensitivity analysis is presented in Figure 1. Only estimated bounds are presented, and issues on inference are omitted as they are not the focus of this paper. On the left end with $\lambda = 0$, monotonicity holds and we obtain the tightest possible bounds. If we relax monotonicity completely, we obtain the bounds on the right end. By varying the sensitivity parameter λ from 0 to 1, we obtain bounds for the intermediate cases.

Environment E1 is the original policy environment (i.e., no extrapolation), acting as our benchmark. At $\lambda = 0$, monotonicity holds, so we are point identified, yielding the original LATE of -0.083. It is evident here that, even without extrapolating, bounds can be very wide when monotonicity does not hold.

Environment E2 has $Pr(CA|C) = 0.05, Pr(NC|N) = 0$: the subsidy hence incentivizes 5% of the compliers (i.e., have a third child if and only if the first two are of the same gender) to become always-takers while none of the existing never-takers becomes compliers. The PRTE here can equivalently be interpreted as the TE of the CC group. The bounds are linear, because we are placing bounds on the TE of a subpopulation of the original complier distribution. The properties

[16] Policy makers are more likely interested in the former rather than the latter question. The latter questions apply more to researchers who are interested in the external validity of their conclusions on the TE.

Figure 1: Plot of $PRTE = E[Y(1) - Y(0)|G \in \{CC, NC\}]$ bounds against λ. Policy Environment E1 has $Pr(CA|C) = 0, Pr(NC|N) = 0$; Policy Environment E2 $Pr(CA|C) = 0.05, Pr(NC|N) = 0$; Policy Environment E3 has $Pr(CA|C) = 0.05, Pr(NC|N) = 0.05$. E4 use the same value for extrapolation parameters as E3 and imposes the additional condition that $\mu_{G1} \leq \mu_{G0}$ for all G. Black line is TSLS benchmark of -0.083.



of the original TEC (e.g., bounds are linear in λ, a result in Noack (2021)) will still hold, and wider bounds are obtained from having a smaller subpopulation. Under monotonicity, the effect of a third child for CC is between -14% and -3.4%.

Environment E3 with $Pr(CA|C) = 0.05, Pr(NC|N) = 0.05$ incentivized both groups. Bounds are much wider when q_{NC} is nonzero, because we have never-takers in the mixture and worst-case bounds of ± 1 are imposed for μ_{NC1}. Linearity in λ is also lost in the presence of the NC group. Under monotonicity, the effect of a third child for CC and NC could be as low as decreasing LFP by 43% and as high as increasing LFP by 20%, and the bounds only get wider when we relax monotonicity.

Since the bounds in E3 are rather uninformative, an additional specification in E4 may be considered. The extrapolation parameters are identical to E3, but now the researcher imposes the additional condition that $\mu_{G1} \leq \mu_{G0}$ for all G. This is reasonable when the researcher believes the the TE for all groups is negative, and the data is only meant to inform how negative this TE can be. Hence, the bounds are uniformly tighter than E3, and the effect is -39% to -2.3% when monotonicity holds. When we relax the sensitivity parameter, the upper bound eventually hits the uninformative bound of 0.

Beyond the baseline sensitivity analysis exercise for PRTE, we can calculate the TE for each of the counterfactual groups, and these may be objects of interest. Namely, NC are those who have a weak preference for gender balance. In the original dataset, they would never have a third child, but once subsidies are available, they have a third child when the first two kids are of the same sex. $E[Y(1) - Y(0)|NC]$ is effect of third child for this subgroup. CC are those who have a strong preference for gender balance, such that childcare subsidies did not incentivize them to have a third kid when their first two kids are of the same sex. DA are those who have a gender preference for two boys or two girls, but their responses are sensitive to childcare subsidies. We can then study the effect of having a third child on labor force participation for these subsets of women.

Overall, it is evident that results can be misleading when the researcher imposes monotonicity when it in fact does not hold. What seemed to be a significantly negative result that is point identified in the original study can instead have uninformative bounds! Similarly, extrapolating the LATE nonparametrically to a different policy environment can yield wide bounds.

6 Conclusion

This paper shows objects of interest, including local average treatment effects (LATE) and policy-relevant treatment effects (PRTE), can be partially identified with a sensitivity parameter that controls the extent monotonicity fails. Without the MTE framework, PRTE is interpreted as the treatment effect for various subpopulations. Identification uses assumptions on proportions of the population that have a particular response to the instrument instead of assumptions on the outcome function. The method presented is useful in many econometric tasks and empirical applications, including those suggested in Mogstad et al. (2018). Econometric applications include: (1) sensitivity analysis of conventional treatment parameters (e.g., LATE, ATT) to monotonicity failure (2) evaluation of the robustness of a study's conclusions to different policy environments and its sensitivity to the presence of defiers (3) testing for independence and exclusion without using a joint test of these assumptions with monotonicity. The threshold crossing extrapolation is useful in environments when the instrument is based on natural experiments (such as gender composition) so individuals affected there are likely different from those affected by a relevant policy. Extrapolation by changing instrument value is useful when the instrument corresponds to a known policy change, but the researcher is interested in the robustness of the result were the policy expanded or contracted.

A Extensions

A.1 Discussion on Sharpness

Suppose Y is discrete for exposition. If Y were continuous, we just take the limit of the discrete sum. Introduce notation h where $h_{g,yy'}$ denotes the proportion of the population that is in treatment response group g, with potential outcomes y when untreated and y' when treated.

$$h_{g,yy'} := Pr(G = g, Y(0) = y, Y(1) = y')$$

This section parameterizes the problem in Equation (8) in terms of point masses h instead of q and μ. By law of total probability, $\sum_{g,y,y'} h_{g,yy'} = 1$. The instrument is discrete and treatment binary

as before. Using the h notation, (q, μ) can be expressed as:

$$\mu_{g0} = \frac{\sum_{y,y'} y h_{g,yy'}}{\sum_{y,y'} h_{g,yy'}} \qquad \mu_{g1} = \frac{\sum_{y,y'} y' h_{g,yy'}}{\sum_{y,y'} h_{g,yy'}} \qquad q_g = \sum_{y,y'} h_{g,yy'}$$

For binary Y, $yy' \in \{00, 01, 10, 11\}$. We have 16 variables. In some cases, the object of interest can be expressed as a linear combination of these point masses i.e., $c'h$, where h is a vector that stacks the point masses. For instance, the ATE $E[Y(1) - Y(0)]$ can be expressed as

$$ATE = h_{A,01} + h_{C,01} + h_{D,01} + h_{N,01} - h_{A,10} - h_{C,10} - h_{D,10} - h_{N,10} \tag{14}$$

The linearity of the ATE is allows Balke and Pearl (1997) to formulate their seminal work using a linear program. However, if we condition on any particular subgroup, then the objective is no longer linear in h. For instance, the LATE is $E[Y(1) - Y(0)|C] = \frac{h_{C01} - h_{C10}}{q_C}$. A similar issue arises with the PRTE. Hence, parameterizing the problem in terms of point masses gives sharp bounds at the cost of losing the linear program.

Constraints are given by the fact that the mass of the variables in each cell must add up to the observable primitives. The analog of Equation (5) for p_{tz} is now:

$$\sum_{y,y',g:T(z)=t} h_{g,yy'} = p_{tz} \tag{15}$$

Instead of having the mean restriction of Equation (2), we now can have equality constraints for the entire distribution. For observed $Pr(T = t, Y = y|Z = z)$,

$$\begin{aligned} Pr(T = 0, Y = y|Z = z) &= \sum_{y',g:T(z)=0} h_{g,yy'} \\ Pr(T = 1, Y = y|Z = z) &= \sum_{y,g:T(z)=1} h_{g,yy'} \end{aligned} \tag{16}$$

In addition, the h's must be in $[0, 1]$. The problem with parameterizing the problem in this way is that the number of parameters grows exponentially as the outcome space grows. If we have a continuous outcome, we will have an infinite-dimensional parameter space.

Parameters (q, H) in Equations (15) and (16) are sufficient to characterize all distributional information. Since we have used all available information in the data in the constraint set, the optimization problem delivers sharp bounds by Theorem 2 of Stoye (2010). Consequently, even without binary outcomes, we can get Noack (2021) bounds for LATE, which are sharp. Running the ATE program gives bounds in Kitagawa (2021).[17]

Remark 6. *(Quantiles). Since the problem can be parameterized in terms of the PDF of various groups, it follows that we can solve for bounds on quantiles of various subgroups similarly using a linear program.*

A.2 PRTE for Multivalued Instrument

With $k > 2$, define $S_{Co} = \{g : g \in S^c_{(z,z')}, \forall z, z' \in \mathcal{Z}\}$ as the set of compliers, and S^*_{Co} the analogous object under the counterfactual policy. Then,

$$\begin{aligned} ATE &= E[Y(1) - Y(0)] \\ LATE &= E[Y(1) - Y(0) | G \in S_{Co}] \\ PRTE &= E[Y(1) - Y(0) | G \in S^*_{Co}] \end{aligned} \tag{17}$$

Constraints and extrapolation parameters are analogous to Section 3. Constraint set can be formed as before, and the extrapolation parameters are $Pr(G^*|S_{G^*})$ for all $G^* \notin \{G : Pr(G|S_G) = 1 \text{ under monotonicity}\}$, where G^* is a group of interest in the PRTE.

To obtain sharp bounds we can parameterize the distribution instead. Index the treatment response as g as before, and index the potential outcomes as $o = (Y(0), Y(1), \cdots)$. Then, the joint distribution can be written as:

$$f(y, t, z) = \int_o \int_g h_{g,o} d\mathcal{G}(T(z) = t) d\mathcal{Y}(Y(t) = y)$$

Here, $\mathcal{G}(T(z) = t)$ denotes the set of groups with $T(z) = t$, and $\mathcal{Y}(Y(t) = y)$ denotes the set of groups with $Y(t) = y$. We might settle for moment conditions that are non-sharp, so we can avoid solving

[17] The parameterization in Equation (16) is analogous to Equation (3) in Kitagawa (2021), which is written in terms of the probability density function.

Table 2: Example of Coefficient Vectors on parameter vector $\mu \in \mathbb{R}^8$

Object of Interest	$c(q)'$
Treatment Effect on Compliers (TEC)	$(0, 0, 1, -1, 0, 0, 0, 0)$
Treatment Effect on Defiers (TED)	$(0, 0, 0, 0, 1, -1, 0, 0)$
Treatment Effect on Marginal Population (TEM)	$(0, 0, q_C, -q_C, q_D, -q_D, 0, 0)/(q_C + q_D)$
Average Treatment Effect (ATE)	$(q_A, -q_A, q_C, -q_C, q_D, -q_D, q_N, -q_N)$
Treatment Effect On Treated (TOT)	$(q_A, -q_A, q_C, -q_C, q_D, -q_D, 0, 0)/(q_C + q_A + q_D)$

an infinite-dimensional problem. As before, the advantage of sharp bounds by parameterizing the problem using the distribution comes as the cost of losing linearity for some objective functions.

B Supplementary Examples

B.1 Binary Instrument

This subsection illustrates the Section 2 using a binary instrument. Using terminology in the literature, the 4 response groups are always-takers (A) with $T_A(0) = T_A(1) = 1$, compliers (C) with $T_C(0) = 0$ and $T_C(1) = 1$, defiers (D) with $T_D(0) = 1$ and $T_D(1) = 0$ and never-takers (N) with $T_N(0) = T_N(1) = 0$. Assumption 2 is equivalent to having no defiers i.e., $q_D = 0$. Under Assumption 1 and Assumption 2, the standard result from Angrist and Imbens (1994) tells us that the two-stage least squares (TSLS) estimand yields $TSLS = \mu_{C1} - \mu_{C0}$, the treatment effect on compliers (often referred to as the local average treatment effect (LATE)). But when there are defiers, the TSLS estimand can no longer be interpreted in the same way.

The vector of proportions and conditional means are:

$$q = (q_A, q_C, q_D, q_N)'$$
$$\mu = (\mu_{A1}, \mu_{A0}, \mu_{C1}, \mu_{C0}, \mu_{D1}, \mu_{D0}, \mu_{N1}, \mu_{N0})'$$

Our objects of interest will take the general form of $c(q)'\mu$, where $c : [0, 1]^4 \to \mathbb{R}^8$. For instance, the average treatment effect (ATE) for the population weights the TE for each group by their relevant proportion, so $c_{ATE}(q) = (q_A, -q_A, q_C, -q_C, q_D, -q_D, q_N, -q_N)'$. More examples are provided in Table 2.

For a given q, we can form constraint set $\mathcal{M}(q)$. An example of an ex ante restriction is the Roy (1951) model, where $\mu_{A1} \geq \mu_{A0}$ and $\mu_{N0} \geq \mu_{N1}$. With minimal restrictions, observed means are:

$$\begin{aligned} q_N\mu_{N0} + q_C\mu_{C0} &= (q_N + q_C)E[Y|T=0, Z=0] \\ q_A\mu_{A1} + q_D\mu_{D1} &= (q_A + q_D)E[Y|T=1, Z=0] \\ q_N\mu_{N0} + q_D\mu_{D0} &= (q_N + q_D)E[Y|T=0, Z=1] \\ q_A\mu_{A1} + q_C\mu_{C1} &= (q_A + q_C)E[Y|T=1, Z=1] \end{aligned} \tag{18}$$

Turning to $\mathcal{Q}(\lambda)$, first observe that once q_D is known, the proportions of all other groups are immediately identified by the data:[18]

$$\begin{aligned} q_A &= 1 - p_{00} - q_D \\ q_N &= 1 - p_{11} - q_D \\ q_C &= p_{00} + p_{11} - 1 + q_D \end{aligned} \tag{19}$$

Consider $q_D/q_C \leq \lambda$, with λ being the sensitivity parameter, telling us the proportion of defiers we allow relative to the mass of compliers. By construction, $\lambda \in [0, 1)$ as $q_D \leq q_C$, so $\lambda = 0$ is where monotonicity holds, and $\lambda \to 1$ places no restrictions on defiers.[19] This setup makes sensitivity comparable across applications: suppose we have $q_D = 0.01$ - if $q_C = 0.5$, then the violation of monotonicity is relatively small; but if $q_C = 0.02$, the violation would be rather large. λ reflects the difference, despite having the same q_D. Consequently, the allowable set for q is:

$$\mathcal{Q}_m(\lambda) = \{q \in [0, 1]^4 : q_D \leq \lambda q_C \text{ and } q \text{ satisfies Equation (19)}\}$$

When monotonicity holds (i.e., q_D=0), $(\mu_{A1}, \mu_{C1}, \mu_{C0}, \mu_{N0})$ are point-identified from the system in Equation (18). Setting $q_D = 0$, since $\mu_{A1} = E[Y|T=1, Z=0]$ and $\mu_{N0} = E[Y|T=0, Z=1]$, by

[18] This follows trivially from solving 3 equations with 3 unknowns:

$$p_{00} = q_N + q_C \qquad p_{11} = q_A + q_C \qquad 1 = q_A + q_C + q_D + q_N$$

[19] Since first stage regression yields $FS = q_C - q_D$, using $\bar{q}_D$ to denote the maximum proportion of defiers, the largest λ that can yield a binding constraint is $\frac{\bar{q}_D}{FS + \bar{q}_D}$.

substitution, $\mu_{C1} - \mu_{C0}$ is point-identified.[20]

Remark 7. *A natural question to ask is how we might know what these q_D are. For sensitivity analysis, λ is simply an assumption that is made. I nonetheless provide two suggestions for how information on q_D can be obtained. (i) Use complementary studies: in Angrist and Evans (1998), we might use some social survey on gender preferences to get some idea of proportion of parents who prefer two girls or two boys. (ii) Use covariates: run the first-stage regression for every subset of covariates W. Since monotonicity requires all first stage coefficients (FS_W) to lie on either side of 0, the mass of observations that have FS_W lie on the opposite side of zero compared to the majority then constitute the defiers. This assumes no sorting by covariates.*

B.2 Additional Constraints

Without making additional assumptions on the primitives of the model, there is one another set of constraints implied by Assumption 1 - the trimming bounds.[21] Let $F_{tz}(y)$ denote the CDF of y conditional on (t, z). Then, denote

$$\begin{aligned} y^L_{z,g}(t) &:= E[Y|T = t, Z = z, Y \leq F^{-1}_{tz}(q_g/p_{tz})] \\ y^U_{z,g}(t) &:= E[Y|T = t, Z = z, Y \geq F^{-1}_{tz}(1 - q_g/p_{tz})] \end{aligned} \tag{20}$$

Conceptually, the observed distribution of $Y|T = t, Z = z$ contains all groups such that $T(z) = t$. For some group g in the mixture, it has mass q_g/p_{tz}, so the lowest possible μ_{gt} is the mean of the bottom tail of that distribution, giving rise to the $y^L_{z,g}(t)$ expression. Similarly, the largest possible μ_{gt} is the mean of the top tail of that distribution, yielding $y^U_{z,g}(t)$.

Then, for all z, g such that $T(z) = t$, trimming bounds take the form:

$$y^L_{z,g}(t) \leq \mu_{gt} \leq y^U_{z,g}(t) \tag{21}$$

[20] To see this,

$$\begin{aligned} \mu_{C1} &= \frac{q_A + q_C}{q_C} E[Y|T = 1, Z = 1] - \frac{q_A}{q_C} E[Y|T = 1, Z = 0] \\ \mu_{C0} &= \frac{q_N + q_C}{q_C} E[Y|T = 0, Z = 0] - \frac{q_N}{q_C} E[Y|T = 0, Z = 1] \end{aligned}$$

[21] Subsequent work like Semenova (2020) call these the Lee Bounds, but I use "trimming" bounds as they are more descriptive of the method, and remains consistent with the term used in Lee (2009).

If we are interested in the means of a subgroup and we only have information on $Y|T, Z$ for a given (T, Z), the trimming bounds will be sharp, a result in Lee (2009).

C Implementation Details

C.1 General Results

Denote the estimate obtained from the sample $(\hat{\underline{\beta}}_\lambda, \hat{\overline{\beta}}_\lambda)$ for the lower and upper bounds respectively for the problem in Equation (7). These estimates are consistent.

Theorem 2. *(Consistency). Suppose Assumption 1 and 3 hold, constraint sets $\mathcal{M}(q)$ and $\mathcal{Q}(\lambda)$ are continuous in the population CDF, and observations are drawn independently from the same population. Then, $\hat{\overline{\beta}}_\lambda \xrightarrow{p} \overline{\beta}_\lambda$ and $\hat{\underline{\beta}}_\lambda \xrightarrow{p} \underline{\beta}_\lambda$.*

Consistency comes from the fact that the objective value is a continuous function of the distribution. We require data to be iid to apply the Glivenko-Cantelli theorem that gives convergence in the empirical CDF. Consistency then follows from continuous mapping theorem after proving continuity in the program.

The sensitivity parameter also has a nice feature stated in Theorem 3, which reduces the number of inequality constraints. Define adjacency in the following manner: Instrument values $z, z' \in \mathcal{Z}$, with $Pr(T(z) = 1) > Pr(T(z') = 1)$ are adjacent if there does not exist $z^* \in \mathcal{Z}$ such that $Pr(T(z) = 1) > Pr(T(z^*) = 1) > Pr(T(z') = 1)$.

Theorem 3. *(Adjacency) Let $z_0, z_1, z, z' \in \mathcal{Z}$. If $\sum_{g \in S^d_{(z_0,z_1)}} q_g \leq \lambda \sum_{g' \in S^c_{(z_0,z_1)}} q_{g'}$ for all adjacent instrument value pairs (z_0, z_1), then $\sum_{g \in S^d_{(z,z')}} q_g \leq \lambda \sum_{g' \in S^c_{(z,z')}} q_{g'}$ for any instrument value pair (z, z').*

This theorem implies that we do not need to consider all instrument pairs - it is sufficient to consider adjacent instrument pairs. For intuition, when there are no defiers both the $(z, z+1)$ and $(z+1, z+2)$ margins, it must be that there are no defiers both the $(z, z+2)$ margin, because the defiers at the $(z, z+2)$ margin must switch at either margin. The advantage of doing this is that we only have to check $k-1$ instead of $\binom{k}{2}$ constraints.

Remark 8. *The property in Theorem 3 is a feature of defining the sensitivity parameter in this way. If we had instead defined the sensitivity parameter an upper bound on the proportion of defiers, we no longer have this property. To see this, suppose we have three discrete instrument values* $\{0, 1, 2\}$*. Sensitivity parameter* η *is such that* $q_{(1,0,0)} + q_{(1,0,1)} \leq \eta^*$ *and* $q_{(0,1,0)} + q_{(1,0,0)} \leq \eta^*$ *at the* $(0, 1)$ *and* $(1, 2)$ *margin of the instrument respectively. In the worst case, we will have* $q_{(1,0,0)} = \eta^*$ *and* $q_{(1,0,0)} = \eta^*$*. Then, at the* $(0, 2)$ *margin,* $q_{(1,0,0)} + q_{(1,0,0)} = 2\eta^*$*, which is not bounded above by* η^**.*

C.2 Implementation Details for Empirical Application

In the empirical application, the dimension of q can be reduced. Although there are 9 response groups in Section 3, we essentially deal with 6 response groups here: (A, CA, CC, D, NC, NC'), where NC' denotes the set of groups that switch from N to anything but a complier in the new policy environment, and D is the original defier group.[22]

The objective function only contains $(q_{CC}, q_{NC}, \mu_{CC}, \mu_{NC})$, so we only have to consider restrictions imposed on those objects. Let $\mathcal{M} = \mathcal{M}_m$, and $\mathcal{Q} = \mathcal{Q}_m$. I split the restrictions in $\mathcal{Q}_m$ and $\mathcal{M}_m$ into three sets. The first set are those on $\mathcal{M}_m$, the second set concerns the p_{tz} restrictions, and the third set concerns the λ restrictions.

The argument for simplification is made for NC', and the one for D is analogous. For any $(q_{NC'}, \mu_{NC'0})$, observe that any combination of $(q_{NA}, q_{ND}, q_{NN}, \mu_{NA}, \mu_{ND}, \mu_{NN})$ that satisfies the system below imposes the same restriction on $(q_{CC}, q_{NC}, \mu_{CC}, \mu_{NC})$ under the first two restriction sets.

$$
\begin{aligned}
\mu_{NC'0} &= \frac{1}{q_{NC'}}[q_{NA}\mu_{NA0} + q_{ND}\mu_{ND0} + q_{NN}\mu_{NN0}] \\
q_{NC'} &= q_{NA} + q_{ND} + q_{NN} \\
\Rightarrow q_N\mu_{N0} &= q_{NA}\mu_{NA0} + q_{NC}\mu_{NC0} + q_{ND}\mu_{ND0} + q_{NN}\mu_{NN0} \\
&= q_{NC}\mu_{NC0} + q_{NC'}\mu_{NC'0}
\end{aligned}
$$

This occurs because $(q_{NA}, q_{ND}, q_{NN}, \mu_{NA}, \mu_{ND}, \mu_{NN})$ imposes restrictions on $(q_{CC}, q_{NC}, \mu_{CC}, \mu_{NC})$ only through the $q_N\mu_{N0}$ term in $\mathcal{M}_m$ and $q_{NC} + q_{NC'} = q_N$ in the p_{tz} restrictions. Consequently, once $(q_{NC'}, \mu_{NC'0})$ is fixed, the optimum $c(q)'\mu$ will be the same regardless of which

[22] To be precise, $q_{NC'} = q_{NA} + q_{ND} + q_{NN}$ and $q_D = q_{DD} + q_{DA}$.

$(q_{NA}, q_{ND}, q_{NN}, \mu_{NA}, \mu_{ND}, \mu_{NN})$ we choose. The same argument can be made for q_D.

However, this is not so with the sensitivity restrictions. The specific choice of q_{DD} and q_{ND} makes restrictions on q_{CC} and q_{NC}. With q_{CC} and q_{NC} as arguments in the optimization problem, the optimum is found by using the least restrictive setting for q_{CC} and q_{NC}. Since the restrictions from the other two sets of constraints do not change regardless of what q_{DD} we choose, we can set $q_{DD} = q_{ND} = 0$ innocuously. Then, $q_{DD} + q_{ND} \leq \lambda(q_{CC} + q_{NC})$ is automatically satisfied. We hence only need to consider $q_D = q_{DD} + q_{DA} \leq \lambda(q_{CC} + q_{CA})$. This will give us the optimum, because the objective value has to perform weakly better with one fewer constraint. Hence, it is sufficient to optimize over 6 groups.

C.3 Inference

Using the reasoning from Appendix C.2, there are essentially six groups when using $\mathcal{M}_m$ and $\mathcal{Q}_m$: (A, CA, CC, D, NC, NC').

Proportion restrictions on p_{tz} yields:

$$\begin{aligned} E[1 - q_D - q_A - (1-T)(1-Z)] &= 0 \\ E[1 - q_D - q_{NC} - q_{NC'} - TZ] &= 0 \\ E[TZ + (1-T)(1-Z) - 1 + q_D - q_{CC} - q_{CA}] &= 0 \end{aligned} \tag{22}$$

Observed means constraints are:

$$\begin{aligned} E\left[\left(\frac{q_{NC}\mu_{NC0} + q_{NC'}\mu_{NC'0} + q_{CC}\mu_{CC0} + q_{CA}\mu_{CA0}}{q_{NC} + q_{NC'} + q_{CC} + q_{CA}} - Y\right)(1-T)(1-Z)\right] &= 0 \\ E\left[\left(\frac{q_A\mu_{A1} + q_D\mu_{D1}}{q_A + q_D} - Y\right)T(1-Z)\right] &= 0 \\ E\left[\left(\frac{q_{NC}\mu_{NC0} + q_{NC'}\mu_{NC'0} + q_D\mu_{D0}}{q_{NC} + q_{NC'} + q_D} - Y\right)(1-T)Z\right] &= 0 \\ E\left[\left(\frac{q_A\mu_{A1} + q_{CA}\mu_{CA1} + q_{CC}\mu_{CC1}}{q_A + q_{CA} + q_{CC}} - Y\right)TZ\right] &= 0 \end{aligned} \tag{23}$$

Finally, there are inequality constraints imposed by a binary outcome, and further restrictions on

the q's imposed by the sensitivity parameter.

$$0 \leq \mu_{gt} \leq 1, \qquad 0 \leq q_g \leq 1, \qquad q_D \leq \lambda(q_{CC} + q_{CA}), \qquad \sum_g q_g = 1 \tag{24}$$

Without additional a priori assumptions, these bounds are sufficient because we have a binary outcome. If the outcome were not binary, we may additionally impose trimming bounds from Appendix B.2. Then, the problem can be written as a set of moment inequalities.

Using the set of moment equalities and inequalities, we can then do inference on (q, μ). In the basic case with only moment equalities, standard GMM can be applied. With moment inequalities asymptotic distribution of these unknowns can be found by using the algorithm in Andrews and Soares (2010) for this set of moment conditions. Then, we can invert the test to form a confidence set. Then, optimize β over the confidence set to obtain a confidence interval for the object of interest.

D Proof of Results

Denote our target object as b, so for a given q, the set of feasible values in the inner problem is:

$$\mathcal{B}(q) = \{b \in \mathbb{R} : b = c(q)'\mu \text{ for some } \mu \in \mathcal{M}(q)\} \tag{25}$$

Lemma 1. *Under Assumption 1, suppose that $\mathcal{M}(q)$ is convex for some fixed q. Then, either $\mathcal{M}(q)$ is empty and hence $\mathcal{B}(q)$ is empty, or the closure of $\mathcal{B}(q)$ is equal to the interval $[\underline{R}(q), \overline{R}(q)]$, defined in Equation (8). Further, if $\mathcal{M}(q)$ can be written as a system of linear inequalities in μ, both optimization problems are linear programs.*

Proof of Lemma 1. Convex $\mathcal{M}(q)$ is either empty or nonempty. If $\mathcal{M}(q)$ is empty, then by definition $\mathcal{B}(q) = \emptyset$. Next consider a nonempty $\mathcal{M}(q)$. Since a linear mapping of a convex set also yields a convex set, and $c(q)'\mu$ is a linear map of μ, it follows that $\mathcal{B}(q)$ is a convex set. Thus, any $b \in [\underline{R}(q), \overline{R}(q)]$ must also be in $\mathcal{B}(q)$. Proving that optimization problems are indeed linear programs is straightforward from its construction. The constraints are linear in μ and the objective function is a linear function of μ. □

Proof of Theorem 1. There are two parts to the theorem. The first concerns convexity and the second concerns sharpness.

Part I

Proof for an empty $\mathcal{B}_\lambda$ is identical to the proof in Lemma 1. Only consider nonempty $\mathcal{M}(q)$. The objective is to show that any $b \in [\underline{\beta}_\lambda, \overline{\beta}_\lambda]$ is achievable for some $q \in \mathcal{Q}(\lambda)$. To do this, I first show first show that $\overline{R}(q)$ and $\underline{R}(q)$ are continuous in q. Continuity of these objects can then be used to complete the argument.

Apply Theorem 2 from Wets (1985) that the objective value of a linear program is continuous in its hyperparameters. The sufficient condition is that the feasible set in both the primal and dual linear programs are continuous in the hyperparameters. For the dual problem, it is assumed that $\mathcal{M}(q)$ is bounded by Assumption 3(a), so Corollary 11 from Wets (1985) implies that the feasible set of the dual problem is continuous in the hyperparameters. Turning to the primal problem, continuity in the hyperparameters is given by Assumption 3(b). The conditions for the Wets (1985) theorem is hence satisfied. Then, using Theorem 2 from Wets (1985), and the fact that the composition of continuous functions is continuous, with $c(q)$ continuous in q due to Assumption 3(c), $\underline{R}(q)$ and $\overline{R}(q)$ are continuous in q.

It remains to show that any $b \in [\underline{\beta}_\lambda, \overline{\beta}_\lambda]$ is achievable for some $q \in \mathcal{Q}(\lambda)$. Let q^0 denote the q value where monotonicity holds. If $\mathcal{M}(q^0)$ is empty, let q^0 be the smallest vector of defier proportions (i.e., where $\max_{(z,z')} \sum_{g:S^d_{(z,z')}} q_g$ is the smallest) and call λ_0 the sensitivity parameter where equality holds. The new $\mathcal{M}(q^0)$ guaranteed to be nonempty because we work in the environment where $\exists q \in \mathcal{Q}(\lambda)$ s.t. $\mathcal{M}(q)$ is nonempty. By construction of the the sensitivity parameter, $\forall \lambda \geq \lambda_0$, using q^0 is feasible. Using Lemma 1, any $r \in [\underline{R}(q^0), \overline{R}(q^0)]$ can be satisfied by some $\mu \in \mathcal{M}(q^0)$. Since an analogous argument can be made for $b \in [\underline{\beta}_\lambda, \underline{R}(q^0)]$, it suffices to show that for all $b \in [\overline{R}(q^0), \overline{\beta}_\lambda]$, there exists some $q \in \mathcal{Q}(\lambda)$ such that $b = \overline{R}(q)$.

Let $\overline{q}$ be the q that achieves $\overline{\beta}_\lambda$ i.e., $\overline{\beta}_\lambda = \overline{R}(\overline{q})$. With slight abuse of notation, let $[q^0, \overline{q}]$ denote the connected set on $\mathbb{R}^{d_q-k-1}$ between q^0 and $\overline{q}$. Since by Assumption 3(d) $\mathcal{Q}(\lambda)$ is convex, any $q \in [q^0, \overline{q}]$ must also lie in $\mathcal{Q}(\lambda)$. Using the fact that the image of a connected set is connected for a continuous mapping, $\overline{R}([q^0, \overline{q}])$ is a connected set. Since $\overline{R}(q^0)$ and $\overline{R}(\overline{q})$ are both feasible, and $\overline{R}(.) \in \mathbb{R}$, $[\overline{R}(q^0), \overline{R}(\overline{q})] \subseteq \overline{R}[q^0, \overline{q}])$. Hence, $\exists q \in [q^0, \overline{q}] \subseteq \mathcal{Q}(\lambda)$ such that $\overline{R}(q) \in [\overline{R}(q^0)), \overline{R}(\overline{q})]$.

Part II

Since $\mathcal{M}_m(q)$ satisfies Assumption 3(b), the conditions of Theorem 1 are satisfied, so the identified set is convex by Theorem 1. This implies that for all $b \in [\underline{\beta}_\lambda, \overline{\beta}_\lambda]$, there exists $q \in \mathcal{Q}(\lambda)$ and $\mu \in \mathcal{M}(q)$ such that $b = c(q)'\mu$. For sharpness, it remains to show that data generated by this pair (q, μ) is permissible by the data i.e., the joint distribution matches the observed distribution, $F(\tilde{T}, \tilde{Z}, \tilde{Y}) = F(T, Z, Y)$, where $F(.)$ is the CDF, (T, Z, Y) is the observed distribution and $(\tilde{T}, \tilde{Z}, \tilde{Y})$ is the constructed distribution.

First consider the restriction in $\mathcal{M}_m(q)$ such that $\sum_{g:T(z)=t} q_g \mu_{gt} = p_{tz} E[Y|T = t, Z = z]$. Since $Y \in \{0, 1\}$, $E[Y|T = t, Z = z] = Pr(Y = 1|T = t, Z = z)$, so the restriction is:

$$p_{tz} Pr(Y = 1|T = t, Z = z) = \sum_{g:T(z)=t} q_g \mu_{gt}$$

Further, there are restrictions that $\sum_{g:T(z)=t} q_g = p_{tz}$ and $\sum_g q_g = 1$ due to restriction that $q \in \mathcal{Q}_m(\lambda)$. These equalities hold by construction for all (t, z) for the given (q, μ).

Observe that the joint distribution of (Y, T, Z) fully characterizes all information in the data. Since we have discrete data, using $f(.)$ to denote the probability mass function, the distribution can be written as its conditionals: $f(Y, T, Z) = f(Y|T, Z)f(T|Z)f(Z)$. It is hence sufficient to show equality for the conditional densities. Given (q, μ), it is immediate that the following equalities hold:

$$Pr(\tilde{T} = t|\tilde{Z} = z) = \sum_{g:T(z)=t} q_g = Pr(T = t|Z = z)$$
$$Pr(\tilde{Y} = 1|\tilde{T} = t, \tilde{Z} = z) = \frac{1}{Pr(\tilde{T} = t|\tilde{Z} = z)} \sum_{g:T(z)=t} q_g \mu_{gt} = Pr(Y = 1|T = t, Z = z)$$

To complete the argument, simply choose

$$Pr(\tilde{Z} = z) = Pr(Z = z)$$

Then, the distributions will be equal, and (q, μ) is permissible by the data.

$\square$

Proof of Theorem 2. Let F_n denote the empirical joint CDF. Using Proposition 3.6.6 of Giné and Nickl (2021), the class of all half-spaces of $\mathbb{R}^d$ is a Vapnik-Cervonenkes (VC) class. Since the joint CDF is the probability measure on the intersection of half spaces, the condition for Glivenko-Cantelli theorem is satisfied. Applying Corollary 3.7.17 in Giné and Nickl (2021), $F_n \xrightarrow{a.s.} F$, where F is the asymptotic joint distribution of (Y, T, Z). This implies $F_n \xrightarrow{p} F$ since almost sure convergence implies convergence in probability.

Consider the $\hat{\overline{\beta}}_\lambda \xrightarrow{p} \overline{\beta}_\lambda$ problem, since the lower bound is analogous. Let $\mathcal{A}(\lambda) := \{(q, \mu) : q \in \mathcal{Q}(\lambda), \mu \in \mathcal{M}(q)\}$. Then, we can rewrite the problem as a single-step optimization over (q, μ):

$$\overline{\beta}_\lambda := \max_{(q,\mu)\in\mathcal{A}(\lambda)} c(q)'\mu$$

By assumption in the theorem, $\mathcal{M}(q)$ and $\mathcal{Q}(\lambda)$ are continuous in F, so $\mathcal{A}(\lambda)$ is continuous in F. By construction, $c(q)'\mu$ is continuous in (q, μ). Applying the theorem of the maximum, $\overline{\beta}_\lambda$ is continuous in F. By continuous mapping theorem, the conclusion follows. $\square$

Lemma 2. *When $\mathcal{Z}^*$ takes $k < \infty$ discrete values, there exists a bijection to a subset of natural numbers, $h : \mathcal{Z}^* \to \mathcal{Z} \subset \mathbb{N}$, such that when $z_1, z_2 \in \mathcal{Z}$, $z_2 > z_1$ implies $Pr(T(z_2) = 1) \geq Pr(T(z_1) = 1)$.*

Proof of Lemma 2. Construct a mapping, then show it is bijective. Let $T^*(z^*)$ denote the potential outcome when given instrument value $z^* \in \mathcal{Z}^*$. For the property to be satisfied in $\mathcal{Z}$, order the values of $\mathcal{Z}^*$ in terms of $Pr(T^*(z^*) = 1)$, then map z^* that yields the smallest $Pr(T^*(z^*) = 1)$ to $z = 0$, the next smallest to $z = 1$ and so on. If there are ties in $Pr(T^*(z^*) = 1)$, it does not matter which z^* value takes the larger z. The minimum exists since $\mathcal{Z}^*$ is finite. Since $\mathcal{Z}^*$ is discrete, this mapping into $\mathbb{N}$ is injective. For surjectivity, simply observe $|\mathcal{Z}^*| = |\mathcal{Z}| = k$. Hence, the mapping is bijective. $\square$

To prove Theorem 3, I first prove a related Lemma.

Lemma 3. *Suppose that for any $z, z' \in \mathcal{Z} \subset \mathbb{N}$, $z > z'$ implies $P(T(z) = 1) \geq P(T(z') = 1)$. For any $n_1, n_2 \in \mathbb{Z}_+$ with $n_2 > n_1$, if $\sum_{g \in S^d_{(z,z+n_1)}} q_g \leq \lambda \sum_{g' \in S^c_{(z,+n_1)}} q_{g'}$ and $\sum_{g \in S^d_{(z+n_1,z+n_2)}} q_g \leq \lambda \sum_{g' \in S^c_{(z+n_1,z+n_2)}} q_{g'}$, then $\sum_{g \in S^d_{(z,z+n_2)}} q_g \leq \lambda \sum_{g' \in S^c_{(z,+n_2)}} q_{g'}$.*

Proof of Lemma 3. The defiers at the $(z, z + n_2)$ margin switch exactly once: either at $(z, z + n_1)$ or $(z + n_1, z + n_2)$. Individuals who switch twice are either always takers or never takers when looking at the $(z, z+n_2)$ margin. It also means that they will be compliers at either one of the two margins and defiers at the other margin. This implies

$$S_2 := S^d_{(z,z+n_2)} \subset S^d_{(z,z+n_1)} \cup S^d_{(z+n_1,z+n_2)} =: S_1$$

To be precise, S_2 consists of defiers who switch exactly once, and $S_1 \backslash S_2$ consists of defiers who switch twice, resulting in their being compliers at one margin.

Let $q(.)$ be the probability measure on sets. By assumption, $q(S^d_{(z,z+n_1)}) \leq \lambda q(S^c_{(z,z+n_1)})$ and $q(S^d_{(z+n_1,z+n_2)}) \leq \lambda q(S^c_{(z+1,z+n_2)})$. Due to binary treatment, the sets $S^d_{(z,z+n_1)}, S^d_{(z+n_1,z+n_2)}$ are disjoint. Similarly, the sets $S^c_{(z,z+n_1)}, S^c_{(z+n_1,z+n_2)}$ are also disjoint. Summing the inequalities,

$$q(S_1) = q(S^d_{(z,z+n_1)}) + q(S^d_{(z+n_1,z+n_2)}) \leq \lambda(q(S^c_{(z,z+n_1)}) + q(S^c_{(z+n_1,z+n_2)}))$$

Consider the set $S^c_{(z,z+n_1)} \cup S^c_{(z+n_1,z+n_2)}$. This set consists of compliers at the $(z, z + n_2)$ margin (which implies $S^c_{(z,z+n_2)}$ is a subset), and $S_1 \backslash S_2$. Namely, $S^c_{(z,z+n_1)} \cup S^c_{(z+n_1,z+n_2)} = (S_1 \backslash S_2) \cup S^c_{(z,z+n_2)}$. Observe that $S^c_{(z,z+n_2)}$ is the set of compliers that switches their treatment status exactly once in the correct direction. Then, the summed inequality is:

$$\begin{aligned}
q(S_2) + (q(S_1) - q(S_2)) &\leq \lambda(q(S^c_{(z,z+n_2)}) + q(S_1) - q(S_2)) \\
&\Rightarrow q(S_2) \leq \lambda q(S^c_{(z,z+n_2)}) - (1 - \lambda)(q(S_1) - q(S_2)) \\
&\Rightarrow q(S^d_{(z,z+n_2)}) \leq \lambda q(S^c_{(z,z+n_2)})
\end{aligned}$$

□

Proof of Theorem 3. Apply Lemma 2 so that we can simply consider the set that satisfies Lemma

3. For $z' > z$, we can write $z' = z + l$ with $l > 0$. Thus, it is sufficient to show that $\sum_{g \in S^d_{(z,z+l)}} q_g \leq \lambda \sum_{g' \in S^c_{(z,z+l)}} q_{g'}$ for any $l \in \mathbb{Z}_+$.

Prove by induction. Apply Lemma 3, using $n_1 = 1, n_2 = 2$. Since $\sum_{g \in S^d_{(z,z+1)}} q_g \leq \lambda \sum_{g' \in S^c_{(z,z+1)}} q_{g'}$ and $\sum_{g \in S^d_{(z+1,z+2)}} q_g \leq \lambda \sum_{g' \in S^c_{(z+1,z+2)}} q_{g'}$, obtain $\sum_{g \in S^d_{(z,z+2)}} q_g \leq \lambda \sum_{g' \in S^c_{(z,z+2)}} q_{g'}$. Suppose $\sum_{g \in S^d_{(z,z+l)}} q_g \leq \lambda \sum_{g' \in S^c_{(z,z+l)}} q_{g'}$, and we want to show $\sum_{g \in S^d_{(z,z+l+1)}} q_g \leq \lambda \sum_{g' \in S^c_{(z,z+l+1)}} q_{g'}$ due to adjacency. Apply Lemma 3 with $n_1 = l, n_2 = l + 1$ to obtain the result. □

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